

Brain MRI Diagnostic Report

VGG16 Transfer Learning · LIME Explainability · Automated Analysis

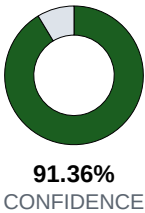
REPORT ID
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Analysis summary

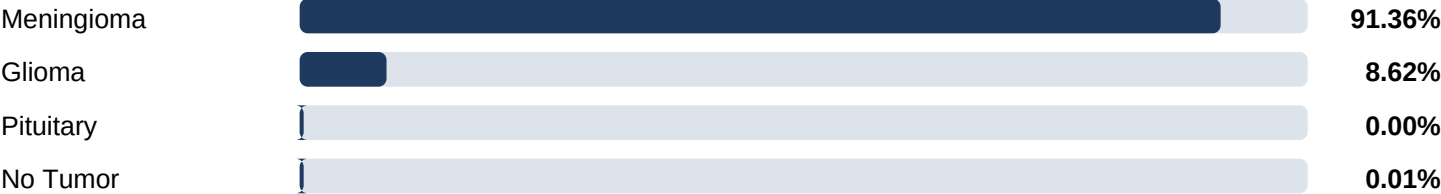
AI model: VGG16 (transfer learning, 4-class head)
Input: Single 2D slice, resized to 128×128 RGB
Pipeline: Inference → occlusion XAI → LIME overlay
Classes: Glioma · Meningioma · Pituitary · No Tumor

Diagnosis

PREDICTION
Meningioma Detected
Raw label: Tumor: meningioma
Top-class confidence: 91.36%

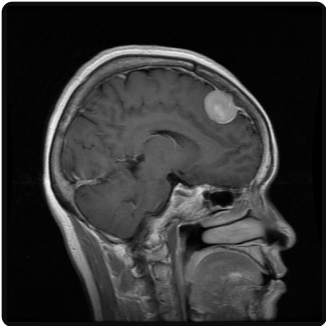


Class probabilities

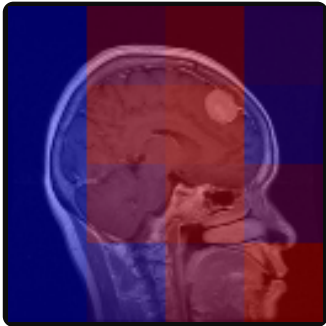


Visualizations

Original MRI



XAI heatmap (model attention)



LIME explanation



Findings & analysis

Model interpretation

Meningiomas are typically extra-axial, dural-based lesions. The model highlights regions consistent with this presentation relative to the learned decision boundary.

LIME explainability summary

LIME perturbs superpixels and fits a local linear model to approximate the classifier near this image. Boundaries mark regions that pushed the score toward the predicted class (Meningioma). Use alongside clinical context.

Report notes

- For research, education, or demonstration only.
- Not FDA-cleared; not a substitute for a radiologist.
- Single-slice uploads omit 3D context and sequence.
- Verify against the full MRI study and patient history.

Technical details

Architecture: VGG16 backbone + dense classifier
Input resolution: 128 × 128 (preprocessed)
Explainability: Occlusion sensitivity map; LIME
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Faizan Ali (FA22-BSCS-187)
M. Mudassar (FA22-BSCS-204)
Supervised by Ms. Fatima Aslam
Department of Computer Science
Lahore Garrison University, Lahore

Not a substitute for professional medical diagnosis
This AI-generated report is a preliminary analysis to assist clinicians and researchers. Final diagnosis and treatment decisions must be made by qualified healthcare professionals based on complete clinical evaluation.